

ALERT-BASED EXECUTION/SOURCE EXECUTION (DCT)

1. OVERVIEW:

What is this use case about?

Answer:

Alert-Based Execution (ABE) proactively identifies and highlights critical supply chain execution issues through real-time alerts across procurement, inbound logistics, and inventory. It enables teams to quickly detect delays, exceptions, and risks, take immediate corrective actions, and safeguard on-time delivery, cost efficiency, and customer satisfaction.

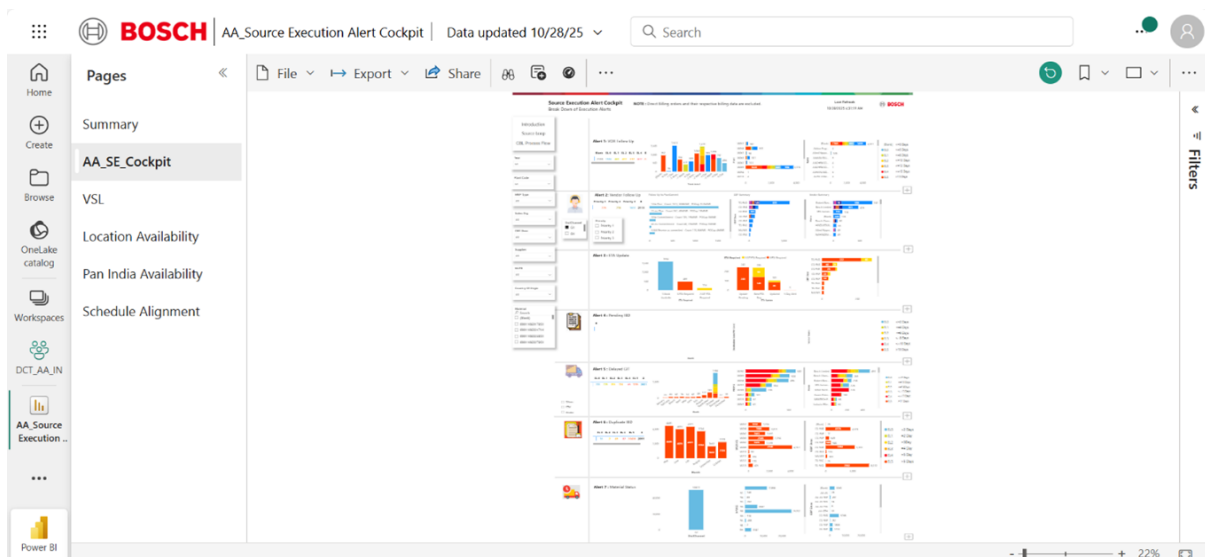
When is the report updated?

Answer:

Refresh - Daily, 7:30 AM IST

2. USER DOCUMENTATION:

This section details the "AA_SE_Cockpit" dashboard, explaining its alerts and filters, which empower users to effectively monitor and manage the execution process.



2.1 Execution Alerts Detail

The dashboard presents seven distinct alerts, each designed to highlight a specific area requiring attention:

2.1.1 Alert 1: VOR Follow Up

Description: This alert track critical orders needing urgent fulfillment, often within 24 hours. "VOR" signifies the product is "vehicle off road" and requires expedited handling.

2.1.2 Alert 2: Vendor Follow Up

Description: Monitors vendors purchase order status, highlighting potential commitment/delivery issues.

2.1.3 Alert 3: ETA Update

Description: This alert focuses on estimated arrival times at Bosch warehouses, tracking stock availability versus requirements and flagging pending or delayed ETAs.

2.1.4 Alert 4: Pending IBD

Description: Identifies deliveries from suppliers that are expected but for which the Inbound Delivery (IBD) process has not yet been completed in the system.

2.1.5 Alert 5: Delayed GIT

Description: Flags goods that are currently in transit from a supplier but have exceeded their expected transit time.

2.1.6 Alert 6: Duplicate IBD

Description: Detects instances where duplicate Inbound Delivery entries might have been created in the system, which can lead to inaccuracies in inventory and planning.

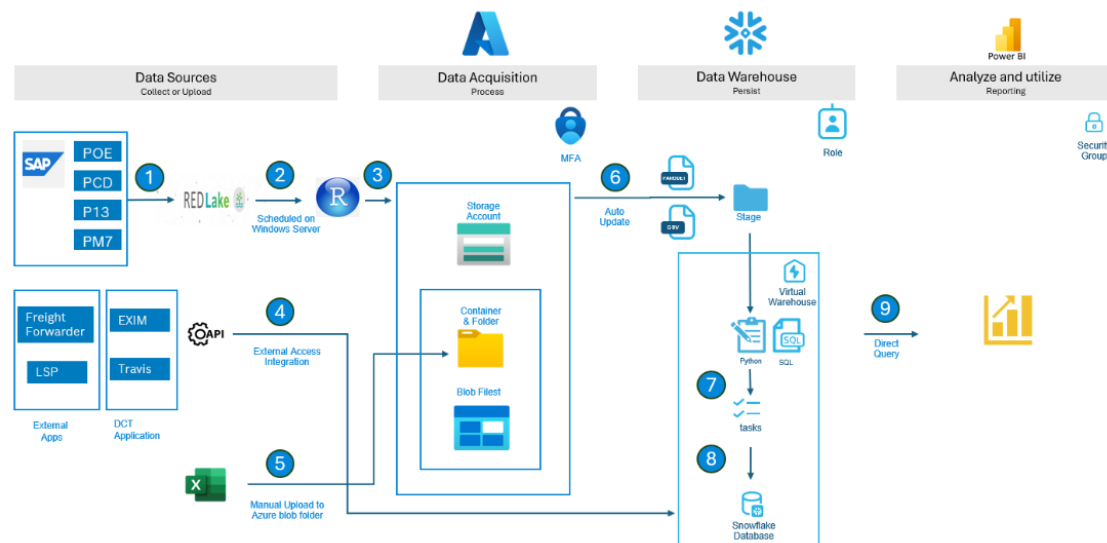
2.1.7 Alert 7: Material Status

Description: Provides an overview of the status of materials (part numbers) within the system, helping to manage inventory and procurement strategies.

3. TECHNICAL DOCUMENTATION:

This section briefly outlines the technical architecture underpinning the Bosch Alert-Based Execution system, highlighting its data flow and key components.

Overseas transportation architecture



3.1 ABE Technical Architecture Overview

The Alert-Based Execution system leverages a robust, cloud-enabled architecture to ensure reliable data acquisition, processing, storage, and reporting. It builds upon established Bosch data infrastructure to deliver actionable insights.

3.2 Key Architectural Phases

3.2.1 Data Sources:

SAP (POE, PCD, PL7): Core enterprise data from various SAP modules, covering procurement, commercial documents, and material management.

External Apps (Freight Forwarder, LSP, Travis, DCT App) & Excel: Integrates data from logistics partners, dedicated CDD applications, and allows for manual supplementary data uploads.

3.2.2 Data Acquisition:

RedLake & Windows Server: Extracts data from SAP via RedLake, with processes scheduled on Windows Servers for automation.

APIs & MFA: External application data is ingested via secure APIs. Multi-Factor Authentication (MFA) ensures secure access during data transfer.

Azure Storage Account: Data is stored in Azure Blob Storage (containers & folders) after initial acquisition, including manual uploads.

Auto Update: Automated processes ensure continuous data refresh within the storage account.

3.2.3 Data Warehouse:

Snowflake (Virtual Warehouse): Data is processed and persistently stored in Snowflake, a cloud data warehousing platform.

Staging: A temporary area for data validation and transformation.

Python & SQL: Used for complex data transformations, cleaning, and orchestration of data pipelines (tasks) within Snowflake.

3.2.4 Analyze & Utilize:

Power BI: The primary reporting tool, connecting directly to Snowflake via Direct Query. This enables near real-time dashboards and reports (like the AA_SE_Cockpit).

Security Groups: Access to Power BI reports is managed through security groups, ensuring data confidentiality and controlled access for authorized users.

This architecture ensures that the ABE system is scalable, secure, and provides timely, accurate data for proactive decision-making.

-> Power BI Dashboard Link: [AA_SE_Cockpit - AA Source Execution Alert Cockpit - Power BI](#)

-> Snowflake Default Role Setup: [Snowflake Default Role Setup - 3S IT - End User Guide - Docupedia](#)

-> How to get Access: [DCT Access.docx](#)

-> Apply for the required role: [oneIDM](#)

QUESTIONS + ANSWERS

1. What is Alert-Based Execution (ABE)?

ANSWER:

Alert-Based Execution (ABE) is a proactive supply chain execution monitoring process that identifies critical operational issues through real-time alerts across procurement, inbound logistics, and inventory operations.

Business Purpose:

- Detect delays, exceptions, and execution risks early
- Enable faster corrective actions
- Improve on-time delivery performance
- Support cost efficiency and customer satisfaction

The system monitors operational exceptions through the AA_SE_Cockpit dashboard.

Source refresh frequency:

- Daily at 7:30 AM IST

2. What is the purpose of the AA_SE_Cockpit dashboard?

ANSWER:

The AA_SE_Cockpit dashboard is used to monitor and manage operational execution alerts related to procurement, inbound logistics, inventory movement, and material status.

Business Purpose:

- Provide visibility into critical execution risks
- Enable proactive follow-up actions
- Improve supply chain responsiveness
- Support inventory and delivery reliability

The dashboard contains seven operational alert categories:

- VOR Follow Up
- Vendor Follow Up
- ETA Update
- Pending IBD
- Delayed GIT
- Duplicate IBD
- Material Status

3. What is VOR Follow Up alert?

ANSWER:

VOR (Vehicle Off Road) Follow Up alert tracks critical orders requiring urgent fulfillment, typically within 24 hours.

Business Purpose:

- Prioritize emergency supply execution
- Prevent vehicle downtime
- Enable expedited operational handling

The alert highlights materials requiring immediate attention due to urgent customer or operational impact.

4. What is Vendor Follow Up alert?

ANSWER:

Vendor Follow Up alert monitors vendor purchase order status and identifies potential vendor commitment or delivery issues.

Business Purpose:

- Detect supplier execution risks early
- Improve vendor follow-up actions
- Prevent supply delays impacting inventory availability

The alert supports proactive coordination with suppliers for pending or delayed commitments.

5. What is ETA Update alert?

ANSWER:

ETA Update alert monitors estimated arrival times of materials to Bosch warehouses.

Business Purpose:

- Track stock availability versus operational requirement
- Identify delayed or pending ETA situations
- Support inbound planning and inventory visibility

The alert helps teams proactively manage supply delays affecting warehouse and inventory operations.

6. What is Pending IBD alert?

ANSWER:

Pending IBD alert identifies supplier deliveries expected by Bosch for which the Inbound Delivery (IBD) process has not yet been completed in the system.

Business Purpose:

- Detect inbound processing gaps
- Improve inbound visibility
- Prevent inventory and planning inaccuracies

The alert supports timely completion of inbound delivery processing activities.

7. What is Delayed GIT alert?

ANSWER:

Delayed GIT alert identifies goods currently in transit from suppliers that have exceeded their expected transit time.

Business Purpose:

- Detect logistics delays
- Improve inbound tracking visibility
- Reduce supply uncertainty impacting inventory planning

The alert helps operational teams proactively follow up on delayed shipments.

8. What is Duplicate IBD alert?

ANSWER:

Duplicate IBD alert detects potential duplicate Inbound Delivery (IBD) entries created in the system.

Business Purpose:

- Prevent inventory inaccuracies
- Avoid planning inconsistencies
- Improve inbound data reliability

Duplicate IBD entries can negatively impact inventory visibility and operational planning.

9. What is Material Status alert in ABE?

ANSWER:

Material Status alert provides visibility into the status of materials (part numbers) within the system.

Business Purpose:

- Support inventory management decisions
- Improve procurement planning
- Enable better material lifecycle visibility

The alert helps teams monitor material-related operational and planning conditions.

10. How frequently is the ABE dashboard refreshed?

ANSWER:

The Alert-Based Execution (ABE) dashboard is refreshed daily at 7:30 AM IST.

This ensures operational teams receive updated execution alerts for procurement, logistics, and inventory monitoring.

11. What are the main data sources used in ABE?

ANSWER:

The Alert-Based Execution system integrates data from multiple operational and enterprise sources.

Primary data sources:

- SAP systems (POE, PCD, PL7)
- Freight Forwarder applications
- LSP systems
- Travis
- DCT App
- Excel uploads

These sources support procurement, logistics, inventory, and execution monitoring processes.

12. What is the role of Snowflake in the ABE architecture?

ANSWER:

Snowflake serves as the cloud data warehouse platform for the Alert-Based Execution system.

Business Purpose:

- Store and process operational execution data
- Support data transformation and orchestration
- Enable scalable reporting and analytics

The architecture uses:

- Staging layer for validation and transformation
- Python and SQL for data processing
- Snowflake Virtual Warehouse for persistent storage

13. What is the role of Power BI in ABE?

ANSWER:

Power BI is the primary reporting and visualization platform used in the Alert-Based Execution system.

Business Purpose:

- Provide operational dashboards and reports
- Enable near real-time monitoring
- Support execution visibility and decision-making

Power BI connects directly to Snowflake using Direct Query.

14. Why is Duplicate IBD dangerous for inventory planning?

ANSWER:

Duplicate IBD entries can create inventory and planning inaccuracies within operational systems.

Operational Impact:

- Incorrect inventory visibility
- Planning inconsistencies
- Misaligned stock availability

- Incorrect inbound tracking

The Duplicate IBD alert helps operational teams identify and resolve such inconsistencies proactively.

15. Why are delayed GIT materials operationally risky?

ANSWER:

Delayed GIT materials increase supply uncertainty because goods remain in transit beyond expected timelines.

Operational Impact:

- Delayed material availability
- Inventory planning disruption
- Potential open order increase
- Reduced operational responsiveness

The Delayed GIT alert enables proactive logistics follow-up and risk mitigation.

16. Required Roles to Access DCT and Snowflake?

ANSWER:

- i. IDM2BCD_SNOWFLAKE_PRXDMLB_BOSCH_DCTINDIA_DEV_GS_DEVELOPER
- ii. IDM2BCD_SNOWFLAKE_PRXDMLB_BOSCH_DCTINDIA_DEV_MA_DEVELOPER